1. **Monte Carlo Integration:** Importance sampling, variance estimation, Riemann and Laplace approximations, applications to model sampling where distribution function is a complicated integral
2. **Markov Chain Monte Carlo (MCMC) methods:** Fundamentals of Markov chains (finite states), transition probabilities, notion of stationarity and ergodicity, stationary distributions, transition kernels, simulated annealing
3. **Monte Carlo methods for Markov chains:** Simulation of ergodic Markov chain data, Metropolis Hastings (MH) algorithm, application to independent data and random walks, Adaptive rejection Metropolis Algorithm (ARMS), Optimization using ARMS
4. **Gibbs Sampler:** Univariate and multivariate Gibbs sampler, comparisons with MH algorithms, slice sampler, hierarchical sampling, two stage Gibbs sampler, hybrid Gibbs samplers.
5. **State Space and Hidden Markov models:** Models for latent processes, Gaussian and non Gaussian models, applications to volatility, mixture population, notion of filtration-Kalman filter and participle filters, estimation and inference for latent states
6. **Multivariate modeling:** Notion of multivariate population, multivariate distribution functions, multivariate normality, principal components, clustering and classification, multivariate analysis of variance
7. **Regression models for time series:** Autoregressive models with exogenous predictors, ARMAX regression, seasonal covariates, estimation of effects, notion of causality and invertibility, problem of simultaneous order and covariate selection.
8. **Model assisted decision making in higher dimensions:** Biasedness and sparsity issues with Lasso, desparsification and bias correction in lasso, high dimensional confidence intervals, computation of debiased lasso with stability properties, extensions to generalised linear (non-Gaussian) models
9. **Modeling heterogeneous data:** Mixture densities, finite and infinite mixture models, simulation and calibration of Gaussian mixtures, finite mixture regression models, clustering, classification and model building for mixture regression models, mixtures of non-Gaussian models.
10. **Supervised learning:** Notion of input and output variables, problem of multiple and noisy inputs, signaling of inputs, covariate selection, deep and embedded

supervised learning, neural networks and k-nearest neighbors, supervised learning in non-Gaussian and time series context

11. **Bayesian model building:** Bayesian estimation of regression coefficients, soft selection and hard selection of covariates, MCMC in Bayesian variable selection, credible intervals, highest posterior density intervals, prediction and predictive accuracy of Bayesian methods.

12. **Model averaging and model ensembling**: Bayesian model averaging, frequentist model averaging, improved confidence intervals and predictive accuracy, averaging of regression models, bagging, boosting and stacking, advantages of model averaging over model selection.

13. **Finite population modeling:** Notion of sample surveys, design and nature of surveys, sampling designs for surveys: Simple random sampling (with and without replacement), stratified random sampling, cluster sampling, systematic sampling, merits and demerits of each design, estimation of the population attributes (average, total) based on the samples from these designs.